Q $y = \text{arccot}(x)$ $\quad \frac{dy}{dx}?$

A $\text{arccot}(x) = y$

$\cot(y) = x$

$\frac{d}{dx} \downarrow$

$-\csc^2(y) \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{-1}{\csc^2(y)}$

Side: $\sin^2 y + \cos^2 y = 1$

$\frac{\sin^2 y}{\sin^2 y} + \frac{\cos^2 y}{\sin^2 y} = \frac{1}{\sin^2 y}$

$1 + \cot^2 y = \csc^2 y$

$\frac{dy}{dx} = \frac{-1}{1 + \cot^2 y}$

$\frac{dy}{dx} = \frac{-1}{1 + x^2}$